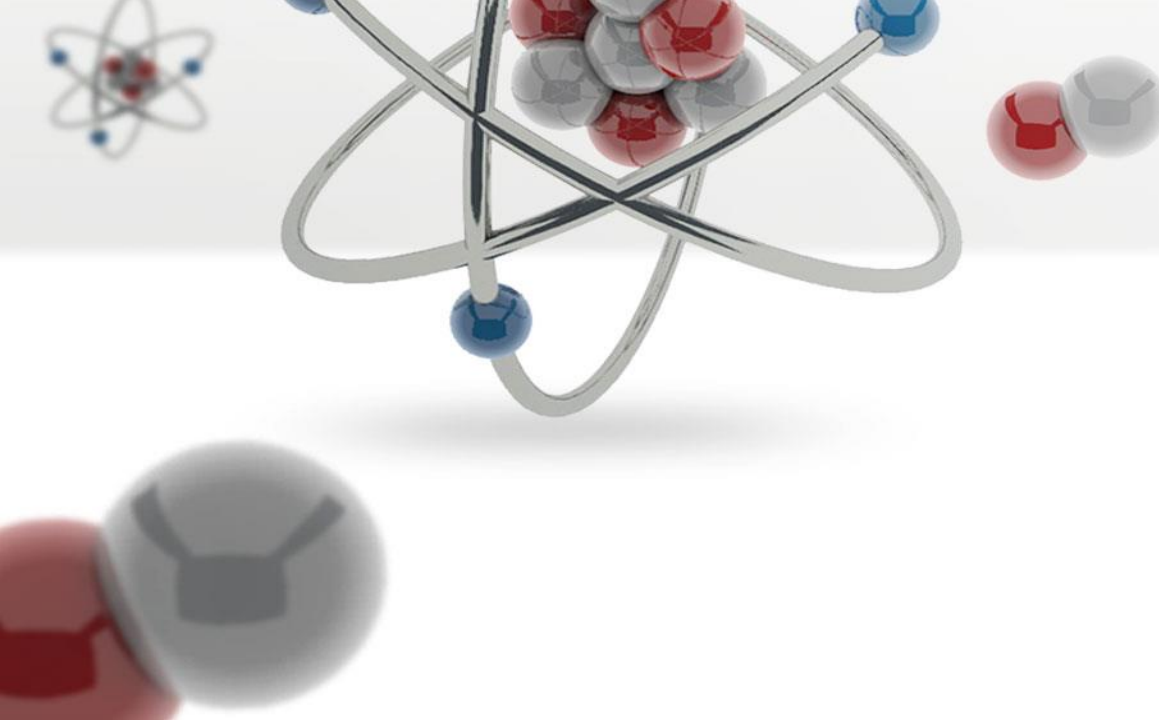




Computational Methods for Quantum Many-Body Systems (CMQMB) - from **artificial atoms** to **high-temperature superconductors**

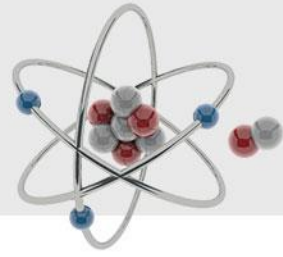
Lecture 5 – Spectral functions and tunneling experiments



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Prof. Dr. Thomas Schäfer

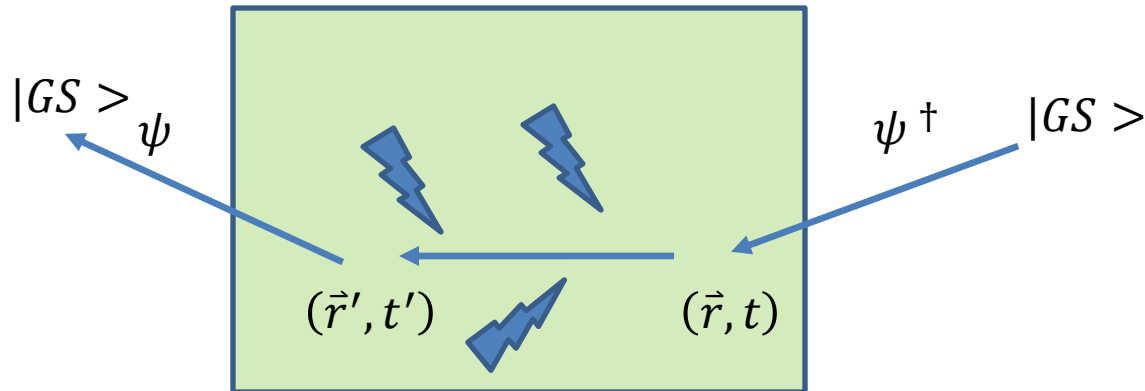
Università degli studi di Trieste (UNITS), Winter Semester 2025



Towards interacting systems...

How can we obtain information from an interacting system?

→ Scattering experiments (theory)

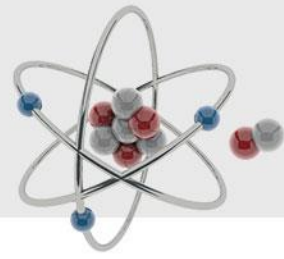


→ Green functions

Three ingredients necessary:

- 1) Many-body treatment → Second quantization ✓
- 2) Time evolution → Pictures of time evolution ✓
- 3) How does the system respond? → Linear response theory ✓

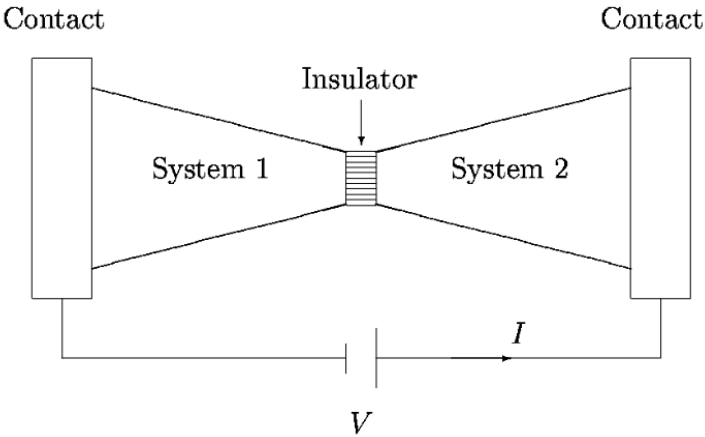
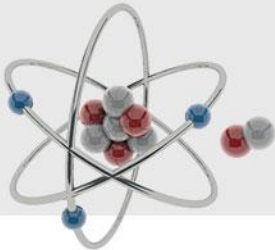
Lecture 11



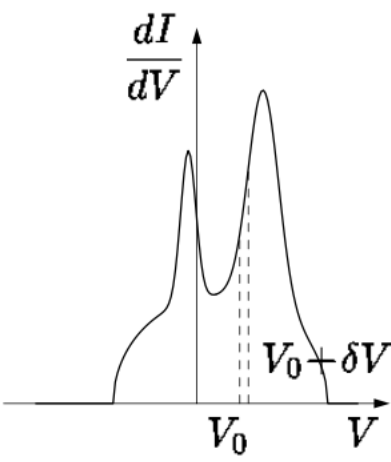
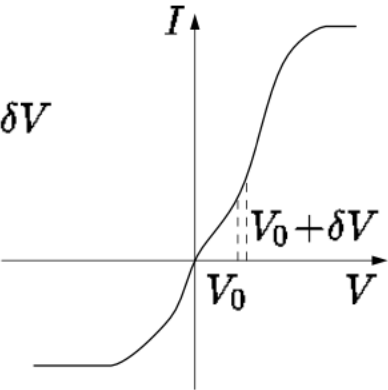
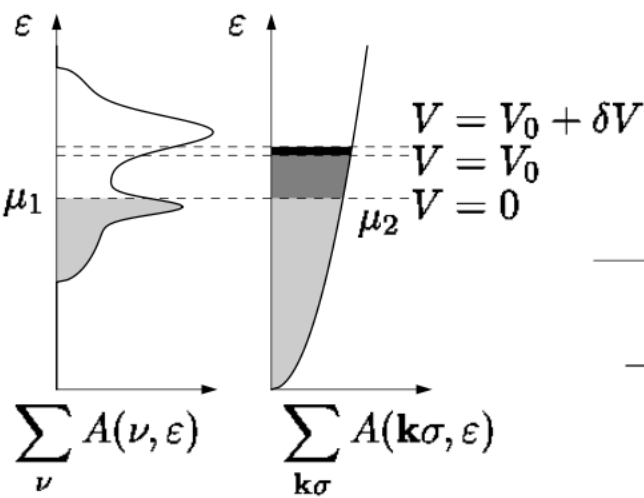
Content and goals

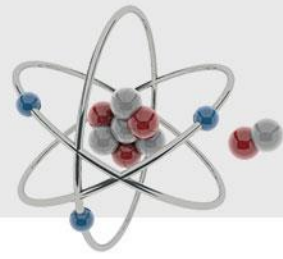
- A closer look at the spectral function
 - Spectral function of the non-interacting system
 - Properties of the spectral function
 - A glimpse towards interactions
- Example: Scanning tunnelling microscopy

Lecture 11: Scanning tunnelling microscopy (STM)

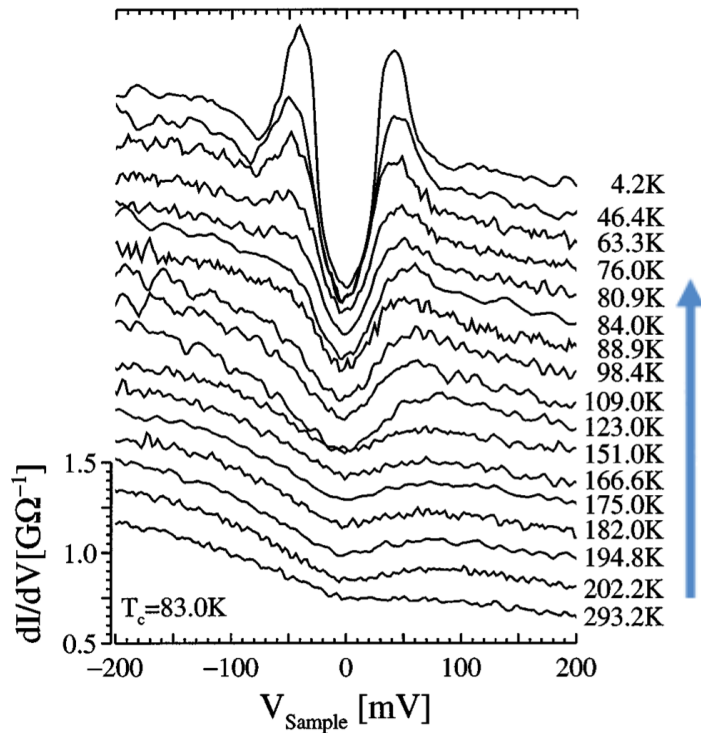


Figures taken from H. Bruus and K. Flensberg, *Introduction to many-body quantum theory in condensed matter physics*

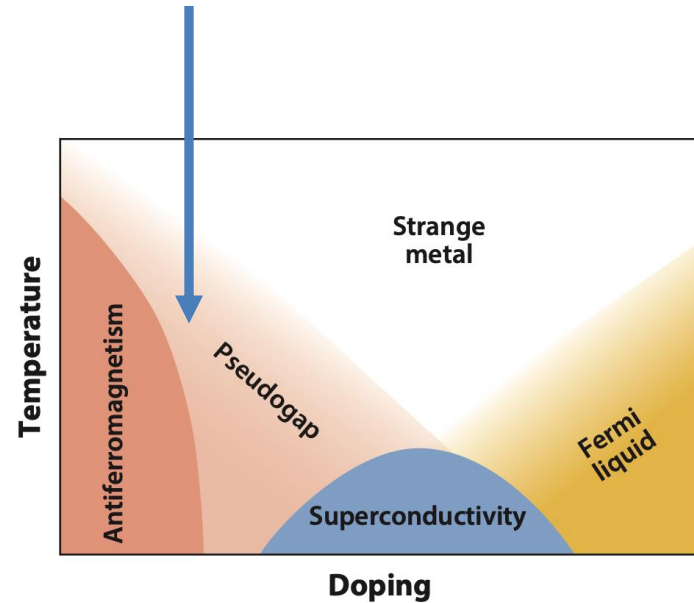




Lecture 11: Scanning tunnelling microscopy (STM)



C. Renner, et al., *Phys. Rev. Lett.* **80**, 149 (1998)



M. Qin, T. Schäfer, et al.,
Annual Review of Condensed Matter Physics **13**, 275-302 (2022)